

SEQUENCE AND SERIES

THEORY

- A sequence is defined as an array of numbers in such a manner so that there is a similarity in a given array, which enables us to determine the term or terms preceding or succeeding to such an array.
- A sequence can be categorized into 3 parts:
 - a) Arithmetic Progression
 - b) Geometric Progression
 - c) Harmonic Progression

	Arithmetic Progression	Geometric Progression
Definition	Series which increases or decreases by a fixed quantity	Series which increases or decreases by a fixed proportion
First Term	a	a
Constant	Common Difference = d	Common Ratio = r
Last Term	$l = t_n = a + (n-1)d$	$l = t_n = a.r^{n-1}$
Sum	$S_n = \frac{n}{2} [2a + (n-1)d]$ $S_n = \frac{n}{2} (a + l)$	$S_n = a \cdot \frac{1-r^n}{1-r}$ when $r < 1$ $S_n = a \cdot \frac{r^n-1}{r-1}$ when $r > 1$

- If three numbers are in G.P., their Logarithms are always in A.P.

Infinite GP Series

$$a + ar + ar^2 + ar^3 + \dots \dots \dots \alpha = \frac{a}{1-r} \quad \text{given } |r| < 1$$

Sum of Natural Numbers:

$$\sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

$$\sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^2(n+1)^2}{4}$$

Harmonic Progression(H.P)

- Three numbers are in H.P, If their reciprocals are in A.P
- a, b, c are in H.P , if $\frac{1}{a} \frac{1}{b} \frac{1}{c}$ are in A.P.
- H.P fails when one of the terms of the A. P is Zero.

$$t_n \text{ of HP} = \frac{1}{t_n \text{ of the corresponding A.P}}$$

Concept of A.M , G.M and H.M

If a & b are any unequal real positive numbers then,

	A.M(A)	G.M(G)	H.M(H)
Definition	$\frac{a+b}{2}$	$+\sqrt{ab}$	$\frac{2ab}{a+b}$
Relation	i) $A >$ ii) $A \times H$	$G >$ $= G^2$	H

Things to remember

- The ratio of the sum of X number of A.Ms to the sum of Y number of A.Ms is always X : Y
- Two numbers can have more than one A.M/G.M/H.M
- A.Ms/G.Ms/ H.Ms are also the members of A.P/G.P/ H.P